

1.3 - StorageCompression - ExamQs.pdf

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IGCSE Computer Science

1.3 Data Storage & Compression

Exam Style Questions

Answer ALL questions and show working where indicated.

Name		Date	
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Score	Percentage	Grade
/ 28		

1

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Question 1

Michele wants to email a file to Elsa. The file is too large so it must be compressed.

- a) Name **two** types of compression that Michele could use.

Compression type 1

Compression type 2

[2]

- b) The file Michele is sending contains the source code for a large computer program. Identify which type of compression would be most suitable for Michele to use. Explain your choice.

Compression type

Explanation

[4]

- c) Elsa wants to email some photos to Michele. Many of the photos are very large files, so Elsa needs to reduce their file size as much as possible. Identify which type of compression would be most suitable for Elsa to use. Explain your choice.

Compression type

Explanation

[4]

2

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Question 2

Miriam needs to use a large high-resolution photo as a thumbnail image on a website. She will use lossy compression to reduce the file size of the photo to create the thumbnail image.

a) State why a smaller file size is appropriate for this situation.

[1]

b) Explain how lossy compression reduces the file size.

[4]

Question 3

The following text is stored as a text file:

She sells sea shells on the seashore.
The shells that she sells are sea shells I am sure.

Explain how lossless compression would compress this file.

[5]

3

Question 4

Computer memory size is measured in multiples of bytes.

Four statements about computer memory sizes are given in the table.

Tick (✓) to show if the statement is True or False.

Statement	True (✓)	False (✓)
25kB is larger than 100 MB		
999MB is larger than 50GB		
3500kB is smaller than 2GB		
2350bytes is smaller than 2kB		

[4]

Question 5

Different units of data can be used to represent the size of a file, as it changes in size.

Fill in the missing units of data, using the list given:

- byte
- gigabyte (GB)
- megabyte (MB)
- nibble

The units of data increase in size from smallest to largest.

Smallest

bit

Kilobyte (kB)

Terabyte (TB)

Largest

[4]

4